

Olympiad Test Paper — Science (Class 7) — Paper 4

Chapter 1: Nutrition in Plants

Paper 4 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	1 — Nutrition in Plants
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic traps: treating respiration as switched off in daylight; reading a net gas measurement as if only one process were running; confusing the role (chlorophyll vs light vs CO₂ vs water) that a single controlled set-up actually isolates; assuming soil nitrogen is a raw material of the day's photosynthesis or that air nitrogen is directly usable; calling a parasite a saprotroph/symbiont; assuming a positive starch test proves the tissue is green; and mixing up xylem (water, up) with phloem (food, out from leaves). Do not write on this paper — mark answers on your answer sheet.

1. A sealed glass chamber holds one bean leaf in bright, steady light. Over a fixed hour the leaf takes in 18 units of carbon dioxide for one process while a second process inside the same leaf releases 6 units of carbon dioxide. A meter on the chamber wall measures only the NET change of the gas in the surrounding air. What does the meter read for carbon dioxide over that hour?

- (A) A net fall of 18 units, because respiration switches off in the light and only photosynthesis runs.
- (B) A net fall of 12 units, because photosynthesis used up more carbon dioxide than respiration released.
- (C) A net rise of 6 units, because only the carbon dioxide given out by the leaf reaches the air.
- (D) No change at all, because the two processes always cancel exactly inside a leaf.

2. In the same chamber as above, the experimenter now wants the carbon dioxide meter to read ZERO net change without removing the leaf or stopping either process. The single condition that achieves this is when:

- (A) the leaf is placed in total darkness so both processes halt and no gas moves.
- (B) the light is made as bright as possible, since maximum photosynthesis balances everything.
- (C) all the chlorophyll is removed so the leaf neither makes nor uses any gas.
- (D) the light is dimmed until photosynthesis takes in carbon dioxide at exactly the same rate respiration releases it.

3. A scientist tracks one carbon atom from the moment it enters a green leaf to the moment it becomes part of a starch grain stored in a potato. Which sequence is correct?

- (A) Carbon dioxide in air → absorbed by roots from soil → carried up in xylem → stored directly as starch.
- (B) Soil minerals → absorbed by roots → carried up in xylem → built into glucose → stored as starch.
- (C) Carbon dioxide in air → enters leaf through stomata → built into glucose in photosynthesis → carried as food in phloem → stored as starch.
- (D) Oxygen in air → enters through stomata → built into glucose in photosynthesis → stored as starch.

4. A pond's green algae gain 40 kilograms of new dry mass over a summer. A student insists 'almost all of that new dry weight must have come from minerals soaked out of the pond mud, since plants feed through their roots.' Choosing the best single correction based on this chapter:

- (A) Most of the new dry mass is water that the algae turned directly into solid plant body.
- (B) Most of the new dry mass came from oxygen the algae absorbed from the pond water.
- (C) The student is right: green plants build their bodies mainly from minerals taken in by their roots.
- (D) Most of the new dry mass is carbon and built from carbon dioxide captured during photosynthesis, not from soil minerals.

5. A destarched plant has three leaves set up for one sunny day, all on the same plant: Leaf 1 is enclosed with a dish of potassium hydroxide (which absorbs carbon dioxide); Leaf 2 has its lower surface coated thickly with grease; Leaf 3 is left untouched. After the day all three are tested with iodine. Which leaves are expected to turn blue-black?

- (A) All three leaves, because they were attached to the same well-lit plant.
- (B) Leaves 1 and 3, because grease cannot stop a leaf from making starch.
- (C) Only Leaf 3, because both Leaf 1 (no carbon dioxide) and Leaf 2 (stomata blocked, so little carbon dioxide enters) were starved of carbon dioxide.
- (D) Only Leaf 1, because potassium hydroxide actually supplies extra carbon dioxide for food-making.

6. A teacher claims a single experiment can prove that BOTH light and chlorophyll are needed for photosynthesis. She takes ONE variegated leaf (green at the edges, white in the centre), covers half of its area — crossing both the green and white zones — with foil, exposes the destarched plant to a sunny day, then does the starch test. Which region turns blue-black, and does the claim hold?

- (A) The entire green border turns blue-black regardless of the foil, showing only chlorophyll matters.
- (B) The whole uncovered half turns blue-black including its white part, showing only light matters.
- (C) Only the part that is both green AND was left uncovered turns blue-black, so the one leaf does demonstrate both requirements.
- (D) No region turns blue-black, because covering any part of a variegated leaf stops all starch-making.

7. A copper beech tree has deep red leaves all summer yet grows tall and healthy in full sun, while a nearby variegated plant's WHITE leaf-patches never test positive for starch. A student asks why the red leaf clearly makes food but the white patch does not. The best explanation is:

- (A) Red pigment is a kind of chlorophyll, so any red or white tissue can photosynthesise equally.
- (B) The red leaf still contains chlorophyll hidden beneath a red pigment, so it photosynthesises; the white patch genuinely lacks chlorophyll, so it cannot.
- (C) The red tree gets its food from the soil, which is why white leaf-patches cannot.
- (D) White tissue makes food using white light while red tissue is purely decorative.

8. An aquatic plant in a beaker of water releases oxygen bubbles in light. A student counts 60 bubbles per minute with a lamp at 10 cm, 15 bubbles per minute at 20 cm, and almost none at 40 cm, keeping water, temperature and the plant the same. The single best conclusion is:

- (A) Moving the lamp away gradually destroys the plant's chlorophyll, so it makes fewer bubbles.
- (B) The rate of photosynthesis falls as the light reaching the plant decreases with distance, confirming light drives the process.
- (C) The plant runs out of water as the lamp moves away, so bubbling slows.
- (D) The bubbles are carbon dioxide, and less of it forms when the lamp is farther off.

9. A young branch on a tree is 'ringed': a complete strip of the outer phloem is peeled away all the way around, while the inner xylem is left intact. The leaves above the ring stay in full sun. Weeks later, what is the most likely observation and why?

- (A) The bark swells just above the ring as sugars pile up, because the leaves still get water through the xylem but the food they make cannot travel down past the broken phloem.
- (B) The leaves wilt and die within a day, because cutting the phloem stops water from reaching them.
- (C) The branch grows faster than ever, because removing phloem frees the leaves to photosynthesise more.
- (D) The leaves immediately stop making food, because the cut bark cuts off their carbon dioxide supply.

10. Surplus glucose in a leaf is rapidly converted to starch for storage. A student asks why the plant bothers to convert it rather than just keeping the glucose. The best reason, consistent with this chapter, is that starch:

- (A) is insoluble, so storing food as starch does not draw extra water into the cells the way dissolved sugar would.
- (B) can be made straight from sunlight without first making glucose.
- (C) is the gas the plant breathes out, so converting glucose helps it respire.
- (D) is the green pigment, so converting glucose to starch greens the leaf.

11. Two leaf discs are taken from the same plant after a sunny day, both genuinely full of starch. Disc P is boiled in water, then in alcohol, then dipped in iodine and turns deep blue-black. Disc Q is dipped straight into iodine with no boiling and barely changes colour. Which single statement best explains BOTH results together?

- (A) Boiling in water created the starch that disc P shows, while disc Q simply never had any.
- (B) Boiling softens the disc and removes its waxy green coat so iodine can reach the starch; the unboiled disc kept iodine out, hiding the starch that was really there.
- (C) Alcohol added starch to disc P, which is why only it reacted with iodine.
- (D) Iodine destroys starch in living tissue, so the boiled, killed disc P was the only one able to react.

12. A child's experiment: destarch a plant, seal one leaf in carbon-dioxide-free air and another in normal air, give a sunny day, and only the normal-air leaf turns blue-black. The child concludes 'carbon dioxide is needed to make starch.' Why was the destarching step at the very start essential for this conclusion to be sound?

- (A) So the leaves would turn green again before the iodine test.
- (B) So that any starch found at the end must have been made during the experiment, not stored from before.
- (C) So carbon dioxide could enter the leaf more easily during the sunny day.
- (D) So iodine would stick to the leaf surface more strongly during testing.

13. A bean seed germinates on damp cotton inside a dark drawer and grows a pale shoot for several days, then stops, yellows and dies. Which statement both EXPLAINS the early

growth and correctly PREDICTS the death?

- (A) The seed lived on food stored inside it; once that ran out, with no light it could not photosynthesise more food, so it died.
- (B) It photosynthesised in the dark using stored chlorophyll, so it should have kept growing forever.
- (C) The damp cotton supplied it with ready-made food, so its death must be due to the cotton drying out.
- (D) It became a parasite on the cotton wool, drawing food until the cotton was used up.

14. *Cuscuta* (a pale, leafless vine) and a mushroom on a dead log are both heterotrophs that lack chlorophyll, yet they are placed in different nutrition groups. A pale *Cuscuta* on a wire fence withers within days, while one on a living shrub thrives; the mushroom flourishes on dead wood and harms nothing living. The classification that fits both facts is:

- (A) *Cuscuta* is a saprotroph living off dead matter such as the wire; the mushroom is a parasite on the log.
- (B) Both are parasites, since both lack chlorophyll and take food from their surroundings.
- (C) *Cuscuta* is a parasite that must take food from a living host; the mushroom is a saprotroph that feeds on dead, decaying matter.
- (D) *Cuscuta* is a symbiont that helps its shrub; the mushroom is an autotroph making food in its cap.

15. A lichen is a fungus and an alga living together: the fungus supplies water, minerals and shelter, the alga supplies food. A student calls this 'a parasite, since the fungus lives off the alga.' The strongest reason the student is wrong is:

- (A) The alga is actually harming the fungus, so the alga is the parasite instead.
- (B) A lichen contains no living organisms, only minerals, so neither can be a parasite.
- (C) Both partners benefit from the arrangement, so it is symbiosis, not parasitism in which one is harmed.
- (D) The fungus makes the food and the alga merely stores it, so neither feeds off the other.

16. Four sealed transparent jars sit in bright light for a day, each with one organism, then the carbon dioxide level inside each is measured: Jar 1 a green fern, Jar 2 a mushroom, Jar 3 a *Cuscuta* vine with no host, Jar 4 a snail. In which jar has the carbon dioxide most likely FALLEN?

- (A) Jar 1, the green fern, because it photosynthesises in the light and uses up carbon dioxide faster than it respire.
- (B) Jar 3, the *Cuscuta* vine, because it photosynthesises strongly using the host it carries.

(C) Jar 2, the mushroom, because fungi absorb carbon dioxide to make food in the light.

(D) Jar 4, the snail, because animals take in carbon dioxide when they breathe.

17. A green pea plant with Rhizobium root-nodules, an insectivorous pitcher plant, and a leafless Cuscuta vine all have a connection to NITROGEN. Which single statement correctly describes how EACH obtains nitrogen?

(A) All three fix nitrogen straight from the air through their green leaves.

(B) All three obtain nitrogen by photosynthesising it from sunlight and water.

(C) The pea gets usable nitrogen via nitrogen-fixing bacteria in its nodules, the pitcher plant gets nitrogen by trapping and digesting insects, and Cuscuta gets nitrogen (with all its food) from its living host.

(D) None of the three needs nitrogen, since plants require only carbon dioxide and water.

18. Air is about 78% nitrogen gas, yet a tomato plant grown in nitrogen-poor SOIL but surrounded by ordinary nitrogen-rich AIR still grows pale and stunted. The reason is that:

(A) the nitrogen in air is a different element from the nitrogen plants need for growth.

(B) leaf stomata are too small to allow any nitrogen gas to pass into the plant.

(C) plants cannot absorb nitrogen gas from the air directly; it must first be fixed into the soil in a usable form.

(D) tomato plants, unlike other plants, do not need nitrogen at all.

19. Two destarched plants are grown for several weeks: plant A in normal soil, plant B in soil sterilised free of nitrogen-fixing bacteria and given no fertiliser. Plant B grows pale and stunted. On one ordinary sunny day a still-greenish leaf from each is tested with iodine. What is observed, and what does it prove about nitrogen?

(A) Only plant A turns blue-black, because nitrogen is one of the raw materials of photosynthesis.

(B) Both leaves can turn blue-black; nitrogen is needed for healthy growth and green colour but is NOT a raw material in the day's photosynthesis.

(C) Neither turns blue-black, because without nitrogen no plant can carry out photosynthesis at all.

(D) Only plant B turns blue-black, because a nitrogen shortage forces the leaf to make extra starch.

20. A teacher claims a plant is an autotroph. Which single observation would most strongly DISPROVE that claim?

(A) It loses water vapour from its leaves on a hot afternoon.

(B) It grows taller and faster when given more sunlight.

(C) It releases streams of oxygen bubbles when brightly lit underwater.

(D) Its tissues contain no chlorophyll and it is seen drawing all of its food from another living plant.

21. A student lists these as autotrophs: (i) a mango tree, (ii) a green pond alga, (iii) a cactus with green stems and no leaves, (iv) a leafless *Cuscuta* vine, (v) blue-green photosynthetic microbes. Exactly ONE entry is wrong. Which, and why?

- (A) (iii) the cactus, because stems can never carry out photosynthesis.
- (B) (ii) the green alga, because algae are too small to make their own food.
- (C) (i) the mango tree, because large trees draw their energy up from the soil.
- (D) (iv) *Cuscuta*, because it lacks chlorophyll and takes ready-made food from a living host, making it a heterotroph.

22. In a food chain rice plant → grasshopper → frog → snake, all the energy flowing through every animal can be traced back to a single original source. That source is:

- (A) the soil minerals absorbed by the rice plant's roots.
- (B) sunlight, captured by the rice plant during photosynthesis.
- (C) the bodies of the animals as they eat one another along the chain.
- (D) the oxygen the animals breathe in as they move along the chain.

23. A student draws snake → frog → grasshopper → green plant with arrows meaning 'is eaten by'. Is the diagram correct for energy flow, and if not, how should it read?

- (A) It is reversed; for 'is eaten by'/energy flow it should read green plant → grasshopper → frog → snake.
- (B) It is already correct, because the top predator should come first in a food chain.
- (C) It should read frog → green plant → snake → grasshopper to balance the chain.
- (D) Food chains have no direction, so either way of drawing it is equally correct.

24. A pond holds green algae, tadpoles that eat the algae, and small fish that eat the tadpoles. A disease suddenly wipes out almost all the green algae. The MOST immediate and direct consequence is:

- (A) the fish increase, because there are now fewer algae competing with the fish for space.
- (B) the tadpoles lose their food supply, so their numbers fall.
- (C) the tadpoles begin making their own food from sunlight to survive.
- (D) the fish die first, before any effect reaches the tadpoles.

25. A balanced sealed terrarium with a green plant survives for weeks in good light. It is then moved into a dark cupboard for two weeks. The most likely outcome is:

- (A) carbon dioxide builds up and oxygen falls, because in the dark the plant only respire, so the plant weakens and may die.
- (B) nothing changes, because the plant simply pauses all its activity in the dark.
- (C) oxygen builds up, because the plant respire faster in the dark than in the light.
- (D) the plant becomes a saprotroph and lives off the soil in the terrarium.

26. A green plant and a small lit candle are sealed together in a jar in BRIGHT LIGHT, and the flame burns noticeably longer than in an identical jar with a candle alone. The

same plant-and-candle jar is then set up in the DARK. Compared with a candle-alone jar in the dark, the flame in the plant-and-candle jar most likely:

- (A) burns even longer, because the plant releases oxygen in the dark just as it does in light.
- (B) burns for exactly the same time, because plants do nothing at all in the dark.
- (C) burns with a green flame, because the plant's chlorophyll colours the smoke.
- (D) goes out sooner, because in the dark the plant stops photosynthesising and instead competes with the flame for oxygen by respiring.

27. A leaf disc kept in the dark for 48 hours, then boiled in water, then in alcohol, then tested with iodine shows NO blue-black colour. A second disc from the same plant after 6 hours of sunlight does turn blue-black. The 48-hour dark step is best described as:

- (A) killing the chlorophyll permanently so the leaf can never make starch again.
- (B) destarching — emptying the leaf of stored starch so that a later test shows only newly made starch.
- (C) slowly adding starch, so the dark disc should actually have tested positive.
- (D) a control proving that iodine simply does not work on leaves kept in the dark.

28. Stomata on a leaf tend to close on a very hot, dry, windy afternoon to save water. A direct consequence for photosynthesis during that afternoon is that it slows because:

- (A) carbon dioxide can no longer enter the leaf in sufficient amounts through the closed pores.
- (B) sunlight can no longer pass through the closed stomata to reach the chlorophyll.
- (C) the chlorophyll inside the leaf is destroyed whenever the stomata close.
- (D) water stops being absorbed by the roots the moment the stomata close.

29. Three leaves from one well-watered plant are tested after a sunny day: leaf 1 normal, leaf 2 has its lower surface coated with a thick layer of grease, leaf 3 normal. Leaf 2 turns far less blue-black than leaves 1 and 3. The best explanation is:

- (A) the grease blocked sunlight from reaching the chlorophyll inside leaf 2.
- (B) the grease dissolved leaf 2's starch before the iodine test could detect it.
- (C) the grease blocked the stomata, cutting off the carbon dioxide supply so leaf 2 made much less starch.
- (D) the grease left an oily film that simply stopped iodine from ever reacting.

30. A leaf on a well-watered plant in still, humid SHADE loses very little water vapour, yet it still makes starch whenever light reaches it. A nearby leaf on a hot, dry, windy day loses water vapour fast. Taken together these observations show that:

- (A) transpiration is required for photosynthesis, so both should always rise and fall together.
- (B) photosynthesis causes transpiration, so a leaf with low water loss cannot have made food.

- (C) water loss and food-making are simply two names for the same single process.
- (D) transpiration and photosynthesis are different processes, so reducing water loss does not stop food-making.

31. A clear plastic bag is tied over a living leaf on a sunny plant and water droplets collect inside; the same is done to a leaf whose cells have been killed, and far fewer droplets form. The best explanation is:

- (A) the living leaf loses water vapour through its working stomata (transpiration); the dead leaf's stomata no longer function.
- (B) the droplets are rainwater that leaked only into the living leaf's bag.
- (C) the living leaf's xylem tubes squirt water directly out of the leaf surface.
- (D) the dead leaf actively makes water inside the bag, which is why it has fewer droplets.

32. A student states: 'Xylem and phloem both carry water in the same direction, just at different speeds.' What is wrong with this?

- (A) Nothing is wrong; both tubes really do carry water upward.
- (B) Both tubes actually carry food rather than water, so only the substance named is wrong.
- (C) Phloem carries food (mostly outward and downward from the leaves), not water; only xylem carries water, and it moves upward.
- (D) Xylem carries food upward while phloem carries water downward.

33. A teacher writes the photosynthesis word equation with ONE deliberate error: 'carbon dioxide + oxygen → (sunlight, chlorophyll) → glucose + water.' Which single correction fixes it?

- (A) Replace 'chlorophyll' with 'soil minerals' as the condition above the arrow.
- (B) Replace 'glucose' on the right with 'starch' as the immediate product.
- (C) Swap the left-hand 'oxygen' for 'water', and the right-hand 'water' for 'oxygen'.
- (D) Replace 'sunlight' above the arrow with 'darkness'.

34. A student writes four statements about photosynthesis; exactly one is INCORRECT. Which?

- (A) Oxygen is one of the raw materials taken in for photosynthesis.
- (B) Chlorophyll captures the light energy used in photosynthesis.
- (C) Glucose is the food made, and some of it is stored as starch.
- (D) Carbon dioxide and water are the raw materials used up.

35. A plant is supplied with carbon dioxide, water, intact chlorophyll and gentle warmth, but kept in TOTAL darkness for a full day; a destarched leaf is then tested with iodine. The result and reason are:

- (A) blue-black, because chlorophyll on its own can make starch even in the dark.
- (B) no blue-black colour, because without light the energy needed for photosynthesis is missing.
- (C) blue-black, because the warmth supplied can replace sunlight as the energy source.
- (D) a red colour, because darkness changes the way iodine reacts with the leaf.

36. A bean plant grows in a sealed clear chamber where carbon dioxide, water and warmth are kept plentiful, but the LIGHT is dimmed week by week. Over the weeks the plant slows and finally weakens. The single best explanation is:

- (A) the plentiful carbon dioxide gradually poisons the plant.
- (B) the plant has used up its chlorophyll because carbon dioxide was kept topped up.
- (C) the plant has turned into a saprotroph that feeds on the chamber walls.
- (D) with too little light the plant cannot photosynthesise enough food to sustain its growth.

37. Two pots of the same plant grow for three weeks: pot 1 in normal soil, pot 2 in soil kept very low in nitrogen. Pot 2's leaves turn pale yellowish and stunted, yet on any single sunny day a still-greenish leaf from pot 2 STILL turns blue-black in the starch test. The combined results show nitrogen is needed for:

- (A) photosynthesis directly, since pot 2 must have made less starch that very day.
- (B) healthy growth and the green colour, but it is NOT a raw material in the day's photosynthesis itself.
- (C) nothing at all, since pot 2 still managed to turn blue-black.
- (D) making the carbon dioxide that the leaf uses inside itself for photosynthesis.

38. Rhizobium living in a legume's root nodules and the legume form a partnership. From the PLANT'S side, what does it GAIN and what does it GIVE?

- (A) It gains usable nitrogen compounds and gives the bacteria food and shelter.
- (B) It gains food (sugars) and gives the bacteria usable nitrogen.
- (C) It gains carbon dioxide and gives the bacteria oxygen.
- (D) It gains water and gives the bacteria chlorophyll.

39. A farmer grows only wheat in the same field every year and yields steadily drop as one mineral is progressively drained. To restore that mineral WITHOUT chemical fertiliser, the best single action is:

- (A) sow more wheat seeds per square metre to make the soil richer in the mineral.
- (B) spray the field with extra water all through the dry months.
- (C) remove every weed, since weeds are the only thing draining the mineral.
- (D) grow a legume crop one season and plough it back, letting Rhizobium in its root nodules fix nitrogen into the soil.

40. Which organism MAKES ITS OWN FOOD, yet is most often studied as part of a partnership with the food-maker rather than on its own?

- (A) The alga in a lichen. (B) The fungus in a lichen.
(C) Rhizobium living in a root nodule. (D) Cuscuta growing on a host plant.

41. A potato tuber, a slice of wheat bread and a piece of egg white are each tested with iodine. The potato and bread turn deep blue-black; the egg white barely changes. The best conclusion is:

- (A) the egg white is the richest source of starch among the three.
(B) iodine only detects starch in plant cells that are still living.
(C) the potato and bread are rich in starch, while the egg white is not.
(D) the blue-black colour shows the potato and bread contain chlorophyll.

42. A leaf's broad, flat, thin shape and the dense packing of green cells near its upper surface together serve one overall purpose. That purpose is to:

- (A) store as much water as possible inside the leaf.
(B) make the leaf heavy enough to resist being blown by wind.
(C) keep insects from being able to land on the leaf.
(D) maximise the capture of sunlight for photosynthesis.

43. An experimenter wants to show that CHLOROPHYLL specifically (not just any leaf tissue) is required for photosynthesis, with the fewest extra variables. Which single set-up does this most directly?

- (A) Keep a whole green plant in the dark for two days, then test a leaf.
(B) Do the starch test on a variegated leaf and compare its green and white parts after a sunny day.
(C) Cover half of a green leaf with foil, expose it to sun, then do the starch test.
(D) Seal a green plant in a jar in light and measure the gas inside.

44. On a variegated leaf (green border, white centre) left in full sun all day, the iodine starch test will turn blue-black:

- (A) only in the green border, where chlorophyll allowed photosynthesis.
(B) only in the white centre, because pale tissue stores the most starch.
(C) over the whole leaf equally, because the leaf shared light evenly.
(D) nowhere on the leaf, because variegated leaves cannot photosynthesise at all.

45. Pick the statement that is FALSE about using the starch test to study photosynthesis.

- (A) A positive starch test on a leaf proves the leaf also contains chlorophyll, since only green parts can ever contain starch.
(B) Iodine turning blue-black indicates the presence of starch.
(C) Destarching the plant first lets you attribute any new starch to the experiment.
(D) Boiling the leaf in alcohol removes chlorophyll so the colour change is easy to see.

46. Why is boiling a leaf in WATER included as a separate step in the starch test, distinct from boiling in alcohol?

- (A) To dissolve out the green chlorophyll, which is the water step's main job.
- (B) To add starch to the leaf before the iodine is applied.
- (C) To release the oxygen trapped inside the leaf so the test can work.
- (D) To soften the leaf and break open its cells so iodine and alcohol can act on it properly later.

47. Comparing nutrition in a green mango tree and a cow, the most FUNDAMENTAL difference is that:

- (A) the cow makes its own food from the grass it eats using sunlight.
- (B) the tree eats tiny soil organisms while the cow eats grass.
- (C) the tree synthesises its food from simple inorganic substances using sunlight, while the cow must take in ready-made organic food.
- (D) both obtain their food ready-made from their surroundings.

48. Four sentences describe modes of nutrition; exactly one is INCORRECT. Which?

- (A) An autotroph builds its food from inorganic raw materials using light.
- (B) A parasite takes food from a living host and harms it.
- (C) A saprotroph attaches to a living host and slowly draws food from it.
- (D) An insectivorous plant photosynthesises and traps insects mainly for nitrogen.

49. A pale, leafless *Cuscuta* vine is found growing over a metal wire fence rather than on a plant, and within days it withers and dies; a *Cuscuta* on a living shrub thrives. This observation best supports the idea that *Cuscuta* is:

- (A) a saprotroph that needs dead matter such as the wire to feed on.
- (B) an autotroph that has only briefly lost its green colour.
- (C) a symbiont that benefits whatever object it grows on.
- (D) a parasite that cannot survive without taking food from a living host.

50. A pitcher plant grows in a boggy, nitrogen-poor place, clearly has bright green leaves, yet still traps and digests insects. A correct conclusion combining BOTH facts is:

- (A) the green colour is merely decorative; it gets all of its food from the insects.
- (B) it photosynthesises in its green leaves for energy-rich food but traps insects to obtain the nitrogen its poor soil lacks.
- (C) it is a parasite, because it feeds on the living insects it traps.
- (D) it is a saprotroph that happens to have evolved green leaves by accident.

51. A sealed transparent box holds a green plant in steady bright light for many hours. The level of which gas would a careful experiment most likely find SLIGHTLY HIGHER at the end than at the start?

- (A) Carbon dioxide.
- (B) Nitrogen.
- (C) Water vapour, which the plant manufactures from sugar.
- (D) Oxygen.

52. Why is a green plant correctly placed at the START of essentially every food chain, while a saprotroph like a mushroom is usually NOT shown as a starting point?

- (A) The mushroom cannot take part in any food chain at all.
- (B) The mushroom makes more food than the plant but releases it only slowly.
- (C) The green plant eats the mushroom to begin the chain.
- (D) The green plant produces new food from sunlight, whereas the saprotroph only breaks down food that already existed.

53. Which statement best explains why nearly ALL animals — even those that eat only other animals — ultimately depend on green plants?

- (A) Green plants directly feed every animal by being eaten by all of them.
- (B) Green plants release most of their oxygen at night, when animals need it most.
- (C) Green plants are the producers that first capture sunlight as food and release the oxygen animals breathe.
- (D) Green plants supply animals with minerals taken straight from the soil.

54. A pond's green algae double in mass over summer while a snail population in the same pond also grows by eating those algae. If, instead, a cloud of fine silt permanently shades the pond so almost no light reaches the algae, the chain of consequences over the following weeks is best described as:

- (A) the snails thrive because shaded algae become softer and easier to eat.
- (B) the algae switch to feeding on the silt and grow even faster in the shade.
- (C) the algae photosynthesise little, so they decline; the snails that feed on them then run short of food and also decline.
- (D) nothing changes, because algae do not actually need light to make food.

55. In a classroom debate a student says: 'Because autotrophs make their own food, they need nothing at all from their surroundings.' The single best correction is:

- (A) Correct — autotrophs are completely independent of their surroundings.
- (B) They actually need to eat small animals to get started each day.
- (C) They take ready-made food from the soil just like heterotrophs do.
- (D) They still need raw materials from their surroundings — carbon dioxide, water and light — to make that food.

56. A destarched plant has one leaf sealed in CO₂-free air (using potassium hydroxide) and another leaf left in normal air; both are on the same plant in bright light for six hours, then tested with iodine. A student writes that 'the leaf in CO₂-free air turned blue-black because removing a gas speeds up food-making.' What is the correct result, and what error did the student make?

- (A) Both leaves turn blue-black, because light alone is enough to make starch.
- (B) Only the normal-air leaf turns blue-black; the student wrongly treated carbon dioxide as a hindrance when it is actually a needed raw material.

(C) Neither leaf turns blue-black, because destarching destroyed their ability to ever make starch.

(D) Only the CO₂-free leaf turns blue-black, exactly as the student predicted, because KOH feeds the leaf.

57. A scientist records a single leaf's net oxygen exchange over 24 hours: net oxygen is released from about 7 a.m. to 6 p.m. and net oxygen is taken IN the rest of the time, with exactly two instants of zero net exchange. What is happening at those two zero-crossing instants?

(A) The leaf has briefly stopped both photosynthesis and respiration.

(B) Photosynthesis and respiration are running at exactly equal rates, so the oxygen made equals the oxygen used.

(C) The leaf is only respiring and is using up all of its own oxygen internally.

(D) Photosynthesis has completely stopped while respiration carries on at full speed.

58. Which single sentence most completely and correctly summarises nutrition in plants as covered in this chapter?

(A) All plants make food only at night, and none ever depend on other organisms.

(B) Plants and animals both make their own food from sunlight in exactly the same way.

(C) Plants take all of their food ready-made from the soil through their roots.

(D) Green plants make their own food by photosynthesis, while some plants and organisms feed in special ways — parasitic, insectivorous, saprotrophic or symbiotic — and soil nitrogen is replenished by nitrogen-fixing bacteria.

59. A girl reasons: 'When a tree grows from a tiny seed into a heavy trunk, all that extra solid wood must have been lifted up by the roots from the soil as solid material.' Using what this chapter teaches about how plants make food, the best correction is:

(A) Most of the added solid material is built from carbon dioxide taken from the air during photosynthesis, not lifted as solid matter from the soil.

(B) Most of the added wood is solid soil drawn up whole through the roots and hardened.

(C) Most of the added wood comes from oxygen the tree absorbed and turned into solid matter.

(D) The girl is right: a tree's whole body is solid soil carried up by the roots over the years.

60. A scientist injects a harmless dye into the water at the base of a plant's stem and, separately, places a labelled sugar at a brightly lit leaf. Hours later the dye is found high in the upper leaves while the labelled sugar is found moving down toward the roots. The two tissues responsible, named in the order (dye, sugar), are:

(A) xylem for the dye, because it carries water upward, and phloem for the sugar, because it carries food away from the leaves.

(B) phloem for the dye and xylem for the sugar, since phloem carries water up and xylem carries food down.

(C) xylem for both, since one set of tubes carries everything up and down the plant.

(D) stomata for the dye and chlorophyll for the sugar, since those are the plant's transport channels.

Solutions — Science (Class 7), Chapter 1:

Nutrition in Plants — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	B	21	D	31	A	41	C	51	D
2	D	12	B	22	B	32	C	42	D	52	D
3	C	13	A	23	A	33	C	43	B	53	C
4	D	14	C	24	B	34	A	44	A	54	C
5	C	15	C	25	A	35	B	45	A	55	D
6	C	16	A	26	D	36	D	46	D	56	B
7	B	17	C	27	B	37	B	47	C	57	B
8	B	18	C	28	A	38	A	48	C	58	D
9	A	19	B	29	C	39	D	49	D	59	A
10	A	20	D	30	D	40	A	50	B	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) Both processes run at once in the lit leaf. Photosynthesis pulls 18 units of CO₂ out of the air; respiration puts 6 units back. The meter sees the NET effect on the air: 18 removed minus 6 added = 12 removed, i.e. a net fall of 12 units. The claim that respiration switches off in light is the day-respiration misconception — respiration never stops, so 18 is wrong. The 'net rise of 6' answer counts only the respiratory output and forgets the much larger photosynthetic uptake from the same air. 'No change/always cancel' is only true at the exact compensation point where the two rates happen to be equal; here uptake clearly exceeds release, so the gases do not cancel.

2. (D) A zero NET reading means CO₂ taken in by photosynthesis equals CO₂ given out by respiration — the compensation point. Lowering light reduces photosynthesis until its uptake rate drops to match the steady respiratory output, giving zero net change while both processes still run. Total darkness does NOT halt both processes: respiration continues, so the meter would show a net RISE, not zero. Maximum brightness drives photosynthesis far above respiration, giving a large net FALL, not balance. Removing

chlorophyll stops photosynthesis entirely while respiration continues, again giving a net rise — and it does stop a process, which the question forbids.

3. (C) The carbon of plant food comes from carbon dioxide, which enters the LEAF through stomata (not the roots/soil), is fixed into glucose during photosynthesis, then travels as food through the PHLOEM to storage organs where surplus glucose is stored as starch. The first wrong option mislocates CO₂ uptake at the roots and uses xylem (which carries water, not food). The minerals option confuses soil minerals with the carbon source — minerals are needed for health but are not the carbon backbone of glucose. The oxygen option names the wrong gas: oxygen is a PRODUCT released, not the raw material that supplies carbon.

4. (D) The bulk of a plant's dry body is made of glucose/starch and related substances whose carbon comes from carbon dioxide fixed in photosynthesis — so the new dry mass is mostly carbon captured from CO₂, not soil minerals. Water is taken in, but it is the source of hydrogen, not the bulk solid dry mass attributed here, and 'turning water directly into solid body' misstates the process. Oxygen is a product RELEASED by photosynthesis, not a building block absorbed to add mass. The student's root-feeding idea is the classic soil-as-food misconception the chapter explicitly overturns: roots supply water and minerals, but food (and thus most dry mass) is made in the leaves from CO₂.

5. (C) Both interventions cut off the carbon dioxide supply, just by different means. Leaf 1 sits in air whose CO₂ is chemically removed by KOH; Leaf 2's stomata — the pores through which CO₂ enters — are clogged by grease on the lower surface where most stomata lie. Without CO₂ neither can make starch, so neither turns blue-black; only the untouched Leaf 3 does. 'All three' ignores that being on a lit plant cannot substitute for the missing raw material. The grease-can't-stop-it option forgets that blocking stomata blocks CO₂ entry, the very thing the starch test here depends on. The KOH-supplies-CO₂ option inverts KOH's role: it ABSORBS carbon dioxide, it does not release it.

6. (C) Starch forms only where ALL requirements are met. Photosynthesis needs chlorophyll (so white tissue fails) and light (so foil-covered tissue fails). The only region satisfying both is green tissue that was also lit — it alone turns blue-black. Comparing the lit-green region with the lit-white region shows chlorophyll is needed; comparing the lit-green region with the covered-green region shows light is needed — so one leaf demonstrates both. 'Entire green border regardless of foil' ignores the light requirement on the shaded green. 'Whole uncovered half including white' ignores the chlorophyll requirement on the white tissue. 'No region at all' is wrong because the lit-green tissue still has everything it needs.

7. (B) The decisive factor is the presence of chlorophyll, the light-capturing pigment. A red leaf only LOOKS non-green because a red pigment masks the chlorophyll that is still present, so it photosynthesises and grows. A white patch contains no chlorophyll at all, so it cannot make food and tests negative for starch — exactly the difference observed. Calling

red pigment 'a kind of chlorophyll' is false; colour alone does not equal chlorophyll. The soil-feeding answer revives the food-from-soil misconception. 'White tissue makes food using white light' is backwards — colourless tissue without chlorophyll cannot photosynthesise no matter the light.

8. (B) Light reaching the plant weakens sharply with distance, and the bubble rate (a measure of photosynthesis) drops in step — strong evidence that light intensity controls the rate of photosynthesis. Only light was varied; water, temperature and the plant were held constant, so those cannot be the cause. Chlorophyll is not destroyed by simply dimming the light; the chlorophyll is intact and works again when the lamp returns. The plant is sitting in water throughout, so it cannot 'run out of water.' The bubbles are oxygen (a product released in light), not carbon dioxide — CO₂ is taken in, not bubbled out, during photosynthesis.

9. (A) Xylem (intact) carries water UP to the leaves, so the leaves keep their water and keep photosynthesising. Phloem carries the FOOD made in the leaves DOWN/out; with the ring broken, that food cannot pass the cut and accumulates as a swelling just above the ring. The wilt-and-die option confuses the two tissues — water travels in the untouched xylem, so water supply is fine. 'Grows faster' is baseless; removing phloem hinders, not helps, the plant. The carbon-dioxide option is irrelevant: CO₂ enters through stomata on the leaves, which the ringing does not touch.

10. (A) Glucose is soluble; large amounts dissolved in cells would upset the cell's water balance by drawing in water. Converting it to insoluble starch lets the plant stockpile food compactly without that osmotic disturbance — which is why surplus glucose is stored as starch. Starch cannot be made directly from sunlight; glucose is made first in photosynthesis, then converted. Starch is a solid food store, not a gas the plant breathes out — that confuses it with carbon dioxide. Starch is not a pigment; the green pigment is chlorophyll, a different substance entirely.

11. (B) Both discs contained starch; the difference is access. Boiling (water then alcohol) softens the tissue, kills the cells and strips the green, waxy chlorophyll layer so iodine can penetrate and reveal the starch in disc P. The raw disc Q's intact waxy surface blocks iodine, so its real starch goes undetected. 'Boiling created the starch' reverses cause and effect — the question states both discs already had starch. 'Alcohol added starch' is false; alcohol removes chlorophyll, it does not add food. 'Iodine destroys starch in living tissue' is invented chemistry — iodine detects starch by turning blue-black, it does not destroy it.

12. (B) Destarching empties the leaves of pre-existing starch. Then any blue-black result at the end is unambiguously NEW starch made during the experiment, so the comparison between the CO₂-free and normal-air leaves fairly tests the effect of carbon dioxide. Without it, leftover starch could give a false positive and ruin the conclusion. Destarching does not 're-green' leaves — colour is unrelated to emptying starch. It does not make CO₂

enter more easily; stomata handle CO₂ regardless of stored starch. It does not improve iodine 'sticking' — that is the job of the later boiling steps, not destarching.

13. (A) A seed contains a store of food that fuels early growth even without light. That explains the initial pale shoot. But photosynthesis needs light; in the dark, once the seed's food store is exhausted the seedling cannot make more, so it weakens and dies — which the option predicts correctly. Photosynthesis cannot occur in the dark, so 'photosynthesised in the dark' is impossible and would not predict death. The cotton supplies water, not food, so it is not a food source. A seedling is a green-capable autotroph-in-waiting, not a parasite drawing food from cotton — cotton is not a living host and offers no food.

14. (C) *Cuscuta* dies on a non-living wire but thrives on a living shrub — proof it needs a LIVING host, the mark of a parasite. The mushroom thrives on DEAD wood and harms nothing alive — the mark of a saprotroph that digests dead, decaying matter. Calling *Cuscuta* a saprotroph contradicts its death on the lifeless wire (it cannot feed on dead matter). Lumping both as parasites ignores that the mushroom needs no living host. The symbiont/autotroph option is doubly wrong: *Cuscuta* harms its host (not mutual benefit), and the mushroom has no chlorophyll, so it cannot be an autotroph making food in its cap.

15. (C) In a parasitic relationship one partner is harmed; in the lichen, the fungus gains food while the alga gains water, minerals and shelter — a mutually beneficial partnership, i.e. symbiosis, not parasitism. The 'alga harms the fungus' option invents harm where the description states mutual benefit. 'No living organisms, only minerals' is false — both the fungus and the alga are living. The role-swap option is factually wrong: the ALGA (with chlorophyll) makes the food, not the fungus, so it misstates who feeds whom and still misses the key point that both benefit.

16. (A) Only an organism that photosynthesises can pull CO₂ out of the air faster than it releases it. The green fern has chlorophyll and in bright light its photosynthesis exceeds its respiration, so CO₂ falls in Jar 1. *Cuscuta* lacks chlorophyll and, with no host, only respire (and starves) — it raises CO₂, and it has no host with it here. The mushroom is a saprotroph with no chlorophyll; it cannot make food in light and only respire, raising CO₂. The snail is an animal that breathes IN oxygen and breathes OUT carbon dioxide, so it raises CO₂, not lowers it.

17. (C) Each uses a different route. The pea cannot use air nitrogen directly; *Rhizobium* in its nodules fixes atmospheric nitrogen into a usable form. The pitcher plant photosynthesises for energy-food but obtains scarce nitrogen by digesting trapped insects. *Cuscuta*, lacking chlorophyll, draws all its nutrition — including nitrogen — from a living host. 'All fix nitrogen through their leaves' is false: ordinary plants cannot fix air nitrogen, and *Cuscuta* is leafless. 'Photosynthesise nitrogen from sunlight' confuses food-making with mineral nutrition — nitrogen is not a product of photosynthesis. 'None needs nitrogen' contradicts the whole point: nitrogen is essential for plant growth.

18. (C) Most plants cannot take up nitrogen gas from the air; usable nitrogen must be supplied through the soil, fixed into a usable form by nitrogen-fixing bacteria (or fertiliser). With poor soil nitrogen, the plant stays pale and stunted despite the air being full of nitrogen gas. The 'different element' option is false — it is the same element nitrogen, just in an unusable gaseous form. The stomata-too-small option invents a size barrier; the real issue is that plants lack the means to use the gas directly, not pore size. 'Tomatoes don't need nitrogen' is false — all plants need nitrogen for healthy green growth.

19. (B) Photosynthesis on a given day needs carbon dioxide, water, light and chlorophyll — NOT soil nitrogen. So as long as plant B's leaf still has some chlorophyll, it can make starch that day and turn blue-black, just like plant A. The long-term paleness and stunting show nitrogen matters for healthy growth and green colour, but the same-day starch test shows it is not a direct raw material of photosynthesis. 'Only A turns blue-black' wrongly treats nitrogen as a photosynthetic raw material. 'Neither' overstates nitrogen's role — a green-ish leaf still photosynthesises. 'Only B, forced to make extra starch' is backwards: nitrogen shortage hampers growth, it does not boost starch-making.

20. (D) An autotroph makes its own food using chlorophyll and light. An organism with NO chlorophyll that takes all its food ready-made from a living host is a parasite, not an autotroph — that directly contradicts the claim. Losing water vapour (transpiration) is normal for almost any leaf and says nothing about how it feeds. Growing faster in more sunlight actually SUPPORTS being an autotroph (more light, more photosynthesis). Releasing oxygen bubbles when lit underwater is a hallmark OF photosynthesis, again supporting the autotroph claim, not disproving it.

21. (D) An autotroph makes its own food. Cuscuta has no chlorophyll and steals ready-made food from a living host — it is a heterotroph (parasite), so listing it as an autotroph is the error. The cactus's green STEMS contain chlorophyll and do photosynthesise (taking over the job of the missing leaves), so (iii) is a true autotroph. Green algae photosynthesise regardless of size, so (ii) is correct — size is irrelevant to being autotrophic. The mango tree makes food in its leaves; it does not 'draw energy from the soil' (that is the soil-as-food misconception), so (i) is a genuine autotroph.

22. (B) Green plants (producers) capture sunlight's energy and store it as food during photosynthesis; every animal in the chain ultimately runs on that captured energy, passed along as one organism eats another. So the original energy source is sunlight. Soil minerals are needed for the plant's health but are not the ENERGY source — the energy comes from light, not minerals. The animals' bodies merely TRANSFER energy that originally came from the plant; they are not the origin. Oxygen is used to release energy from food during respiration, but it is not itself the source of the energy — the food (built from sunlight) is.

23. (A) An 'is eaten by' / energy-flow arrow points from the organism being eaten to the one that eats it, starting at the producer. Energy enters at the green plant and flows up: green plant → grasshopper → frog → snake. The student's chain runs the wrong way.

'Already correct, predator first' inverts the meaning of the arrows. The frog-first jumbled order breaks the actual eating relationships. 'No direction' is false — food chains have a definite direction of energy flow, always beginning with the producer.

24. (B) The algae are the producers feeding the tadpoles directly. Remove the algae and the tadpoles (the next link) immediately lose their food, so their numbers fall first — the most direct consequence. The fish-increase option is wrong: algae and fish do not compete for the same food (the fish eat tadpoles), and losing the base of the chain harms the whole chain, not helps the fish. Tadpoles are animals; they cannot suddenly photosynthesise their own food. The fish are one step further up the chain, so they are affected AFTER the tadpoles decline, not before — the impact travels up from the base.

25. (A) In the light, the terrarium balanced because photosynthesis recycled the CO₂ from respiration and respiration recycled the oxygen from photosynthesis. In the dark, photosynthesis stops but respiration continues: the plant keeps taking in oxygen and giving out CO₂ with nothing to recycle them, so CO₂ rises, oxygen falls, and the plant weakens. 'Nothing changes' assumes the plant halts all activity in the dark — but respiration never stops. 'Oxygen builds up' is impossible: respiration USES oxygen, and there is no photosynthesis to replace it. A green plant does not convert into a saprotroph; it cannot 'live off the soil' for food.

26. (D) In light, the plant's photosynthesis releases oxygen, which feeds the flame and prolongs burning. In the dark there is no photosynthesis, so no oxygen is added; meanwhile the plant respire, consuming oxygen, so it competes with the flame and the flame goes out sooner. 'Burns even longer' wrongly assumes the plant releases oxygen in the dark — photosynthesis needs light. 'Exactly the same' assumes the plant does nothing in the dark, but it respire and uses oxygen. The green-flame answer is nonsense chemistry; chlorophyll does not colour a flame.

27. (B) Two days in the dark stops photosynthesis and lets the leaf use up its stored starch — destarching. The dark disc then tests negative (no starch left), while the lit disc makes fresh starch and tests positive. This is exactly what destarching is for: a clean baseline so any later positive is NEW starch. The dark does not destroy chlorophyll permanently — the leaf makes starch again after 6 hours of light, proving its chlorophyll is intact and working. The dark adds no starch; it removes it, so 'should have tested positive' is wrong. Iodine works fine on any leaf with starch — the dark disc tested negative because it had no starch, not because iodine failed.

28. (A) Stomata are the pores through which carbon dioxide enters the leaf. When they close to conserve water, CO₂ supply drops, and since CO₂ is a raw material, photosynthesis slows. Sunlight does not enter through the stomata — light passes through the leaf surface regardless of whether the tiny pores are open, so 'sunlight blocked' is wrong. Closing stomata does not destroy chlorophyll; the pigment is unaffected. Root water absorption is not switched off by stomatal closure — closing reduces water LOSS, it

does not stop uptake at the roots; in any case the limiting factor for photosynthesis here is the cut-off CO₂.

29. (C) Most stomata are on the lower surface; greasing it clogs them, cutting off the CO₂ needed for photosynthesis, so leaf 2 makes little starch and barely turns blue-black. A thin coat of grease on the underside does not block the sunlight that reaches the upper surface and chlorophyll. Grease does not dissolve starch — starch is unaffected by an oily film. The iodine-blocking idea is wrong: the boiling steps clean the leaf before testing, and in any case the controlled comparison with the normal leaves points to a real reduction in starch made, not a testing artefact.

30. (D) The shaded leaf loses little water (low transpiration) but still makes starch in light (photosynthesis works) — so the two can vary independently and are distinct processes. The 'required/always together' option is contradicted by the very observation: low transpiration with successful food-making. 'Photosynthesis causes transpiration, so low loss means no food' is disproven because the low-loss leaf DID make starch. They are not the same process under two names — transpiration is water vapour leaving the stomata, photosynthesis is food being made; the data show one can be low while the other proceeds.

31. (A) Transpiration is the loss of water vapour through stomata, which then condenses as droplets in the bag. A living leaf's stomata work, so it transpires and droplets form; a dead leaf's cells no longer regulate or drive this loss, so far fewer droplets appear. The rain idea is excluded — the bag is sealed and identical for both, so any difference is the leaf itself. Xylem carries water inside the plant but does not 'squirt' water out of the surface; loss is as vapour through stomata. The dead-leaf option is backwards: a dead leaf does not MAKE water; it loses less because its transpiration has effectively stopped.

32. (C) Xylem carries water (and dissolved minerals) UP from the roots; phloem carries the FOOD made in the leaves outward, mostly downward, to the rest of the plant. So the two tissues carry different substances in different directions — the student is doubly wrong. 'Nothing wrong, both carry water up' repeats the error. 'Both carry food' is wrong because xylem carries water, not food. The role-swap option ('xylem food up, phloem water down') reverses both tissues' jobs. Only the correct option keeps xylem = water/up and phloem = food/out.

33. (C) The correct word equation is: carbon dioxide + water → (sunlight, chlorophyll) → glucose + oxygen. The teacher's version wrongly listed oxygen as a raw material on the left and water as a product on the right. Swapping them — water in as a raw material, oxygen out as a product — fixes it. Chlorophyll (not soil minerals) is the required pigment condition, so that swap is wrong. The immediate food made is glucose, not starch (starch is the later storage form), so that swap is wrong. Photosynthesis needs sunlight, not darkness, so replacing sunlight with darkness is wrong.

34. (A) Oxygen is a PRODUCT released by photosynthesis, not a raw material taken in — that statement is the incorrect one. Chlorophyll does capture the light energy, which is correct. Glucose is indeed the food made, with surplus stored as starch — correct. Carbon dioxide and water are the two raw materials used up — correct. Only the oxygen-as-raw-material claim reverses a product for a reactant.

35. (B) Photosynthesis requires light as its energy source. With CO₂, water and chlorophyll present but NO light, no food is made, so the destarched leaf stays starch-free and does not turn blue-black. Chlorophyll captures light but cannot manufacture starch in the dark on its own — light is essential. Warmth is not the energy source for photosynthesis; it cannot substitute for light. Darkness does not alter iodine chemistry — iodine still turns blue-black with starch and shows nothing without it; here there simply is no starch.

36. (D) Light is the energy source for photosynthesis. As light falls, the plant makes less and less food despite ample CO₂, water and warmth, so it slows and weakens — light is the limiting factor. Plentiful carbon dioxide does not poison a plant; CO₂ is a raw material it uses. Keeping CO₂ topped up does not consume chlorophyll — the two are unrelated. A green plant does not transform into a saprotroph feeding on chamber walls; that is not a real mode by which a green plant lives.

37. (B) The same-day starch test is positive for both pots, showing the day's photosynthesis (needing CO₂, water, light, chlorophyll) does not directly require soil nitrogen. The long-term paleness and stunting in pot 2 show nitrogen is essential for healthy growth and green colour. So nitrogen matters for growth/colour but is not a direct photosynthetic raw material. 'Directly for photosynthesis' is contradicted by pot 2's positive same-day test. 'Nothing at all' ignores the obvious harm to pot 2's growth and colour. Leaves do not 'make their own carbon dioxide' from nitrogen — CO₂ enters from the air through stomata.

38. (A) Rhizobium fixes atmospheric nitrogen into a form the plant can use, so the plant GAINS usable nitrogen. In return the plant, a food-maker, GIVES the bacteria food (sugars) and a home (shelter) in its nodules. The first wrong option reverses the exchange — it is the BACTERIA that supply nitrogen and the plant that supplies food, not the other way round. The plant does not depend on the bacteria for carbon dioxide (it gets CO₂ from air) nor does it hand them oxygen as the deal. The plant does not 'give chlorophyll' — chlorophyll is not a transferable gift, and the plant's gain here is nitrogen, not water.

39. (D) Repeated wheat drains soil nitrogen. Growing a legume (whose Rhizobium nodules fix atmospheric nitrogen) and ploughing it back restores nitrogen naturally, without chemicals — the standard crop-rotation remedy in this chapter. Sowing MORE wheat would drain nitrogen faster, not restore it. Extra water does not add nitrogen to the soil. Removing weeds does not replace the lost mineral; weeds are not the cause of the steady drain — the repeated wheat crop is.

40. (A) The alga in a lichen has chlorophyll and makes the food for the partnership — it is the autotroph (food-maker), studied together with its fungal partner. The fungus in the lichen does NOT make food; it supplies water, minerals and shelter — it is the heterotroph partner, not the food-maker. Rhizobium does not make its own food by photosynthesis; it fixes nitrogen and takes food from the legume. Cuscuta lacks chlorophyll and steals food from a host — it makes no food at all. Only the alga both makes its own food and is studied within a partnership.

41. (C) Iodine turns blue-black in the presence of starch. The potato and bread (both plant storage foods) react strongly, so they are starch-rich; the egg white barely changes, so it is not. The 'egg white richest in starch' option contradicts its weak reaction. 'Only living plant cells' is false — iodine detects starch in dead/cooked tissue too (bread and boiled potato react well). The blue-black colour indicates STARCH, not chlorophyll; chlorophyll is the green pigment and is unrelated to the iodine-starch reaction.

42. (D) A broad, flat leaf spreads a large area to the sun, and the green (chlorophyll-rich) cells packed near the upper, light-facing surface catch that light — both features maximise sunlight capture for photosynthesis. A thin, flat leaf is not built to store large amounts of water (water-storing organs are thick/fleshy). The broad shape makes leaves catch wind, not resist it, so 'heavy to resist wind' is wrong. Leaf shape is not an anti-insect adaptation; its design is about light capture for food-making.

43. (B) A variegated leaf has green (chlorophyll) and white (no chlorophyll) areas on the SAME leaf under the same light, water and air. After a sunny day only the green parts turn blue-black, isolating chlorophyll as the needed factor with minimal extra variables. Keeping a plant in the dark tests the LIGHT requirement, not chlorophyll. Covering half a leaf with foil also tests LIGHT (lit vs shaded), not chlorophyll. Sealing a plant in a jar and measuring gas demonstrates gas exchange, not the specific need for chlorophyll. Only the variegated-leaf comparison directly isolates chlorophyll.

44. (A) Starch forms only where chlorophyll is present to drive photosynthesis. The green border has chlorophyll and makes starch, turning blue-black; the white centre lacks chlorophyll and stays brown. 'Only the white centre' is backwards — pale, chlorophyll-free tissue makes no starch. 'Whole leaf equally' ignores that the white parts cannot photosynthesise even with light. 'Nowhere at all' is wrong because the green parts certainly do photosynthesise and make starch.

45. (A) A positive starch test shows starch is PRESENT, but starch can be present in tissues that are not green and have no chlorophyll — for example a white potato tuber turns blue-black yet has no chlorophyll. So the claim that a positive test proves chlorophyll is the FALSE one. Iodine turning blue-black does indicate starch — true. Destarching first does let new starch be attributed to the experiment — true. Boiling in alcohol removes chlorophyll to make the colour change visible — true. Only the chlorophyll-implication statement is false.

46. (D) Boiling in water softens the leaf and breaks its cells, making the tissue permeable so the later alcohol (chlorophyll removal) and iodine (starch detection) steps work. Dissolving out chlorophyll is the ALCOHOL step's job, not the water step — that is the classic mix-up this question targets. The water step does not ADD starch; nothing in the test adds food. Releasing trapped oxygen is not the purpose; the test detects starch, and the water boil's role is to soften and break the cells.

47. (C) The core difference between autotrophic and heterotrophic nutrition is that the tree (autotroph) builds its own food from inorganic raw materials (CO₂, water) using sunlight, while the cow (heterotroph) cannot make food and must eat ready-made organic food. The cow does NOT photosynthesise its food from grass — it digests grass; only plants make food from sunlight. The tree does not 'eat soil organisms' — it makes its own food, not by eating. 'Both obtain ready-made food' is false for the tree, which manufactures its own.

48. (C) A saprotroph feeds on DEAD and decaying matter, not on a living host — the sentence describing it as attaching to a living host actually describes a PARASITE, so it is the incorrect statement. An autotroph does build food from inorganic raw materials using light — correct. A parasite does take food from a living host and harm it — correct. An insectivorous plant does photosynthesise and trap insects mainly for nitrogen — correct. Only the saprotroph sentence is wrong.

49. (D) Cuscuta thrives only on a LIVING host and dies on a lifeless wire — exactly what a parasite (which feeds on a living host) would do. A saprotroph feeds on dead, decaying matter; a metal wire is neither food nor decaying matter, and a saprotroph would not depend on a living shrub, so that is wrong. An autotroph makes its own food and would not die for lack of a host; Cuscuta is pale/leafless because it genuinely lacks chlorophyll, not temporarily. A symbiont benefits its partner mutually, but Cuscuta harms its host and gains nothing from a wire — not symbiosis.

50. (B) Both facts must be honoured: green leaves mean it photosynthesises (making its energy-rich food), and insect-trapping supplies the scarce nitrogen its poor soil cannot. So it is a green, food-making plant that supplements nitrogen by catching insects. 'Green is decorative, food from insects' ignores that green leaves genuinely photosynthesise. It is not a parasite — a parasite draws food from a LIVING host plant over time; the pitcher plant makes its own food and only uses insects for nitrogen. It is not a saprotroph (which feeds on dead, decaying matter) — its green leaves are functional, not accidental.

51. (D) In steady bright light the plant's photosynthesis outpaces its respiration: it releases oxygen (a product) and consumes carbon dioxide. So oxygen is the gas that ends up slightly higher. Carbon dioxide would be slightly LOWER, since net photosynthesis uses it up. Nitrogen is not taken in or released by photosynthesis or respiration, so it stays essentially unchanged. Water vapour is not 'manufactured from sugar' — water is a raw material the plant uses, and any change in vapour is from transpiration, not the gas the

question's logic singles out as rising from food-making; the clean answer driven by photosynthesis is oxygen.

52. (D) Producers begin food chains because they make brand-new food from sunlight, introducing fresh energy into the chain. A saprotroph like a mushroom makes no new food; it only breaks down already-existing dead matter, so it is not a starting producer. 'Cannot take part at all' is too strong — saprotrophs do feature in food webs as decomposers, just not as the producer start point. The mushroom does NOT make food (it has no chlorophyll), so 'makes more food than the plant' is false. Green plants do not eat mushrooms to begin chains — plants make their own food and are not consumers.

53. (C) Green plants are producers: they capture sunlight as food that flows up every food chain to all animals (even meat-eaters depend on plant-eaters), and they release the oxygen animals need to release that energy by respiration. That is the deep, universal dependence. Not every animal eats plants directly — carnivores eat other animals — so 'eaten by all of them' is false; the dependence is via the chain. Plants make most oxygen in the LIGHT (day) when they photosynthesise, not at night. Plants do not pass soil minerals straight to animals; the key dependence is on plant-made food and plant-released oxygen.

54. (C) Light is the energy source for the algae's photosynthesis. Persistent shade starves them of light, so they make little food and decline (producers fail first). The snails, which depend on the algae for food, then run short and decline too — the impact passes up the chain from the base. 'Snails thrive on softer algae' invents a benefit; less light means less algae, not better food. Algae cannot 'feed on silt' — they are autotrophs that need light, not particles, to make food. 'Algae don't need light' is flatly false; without light photosynthesis stops.

55. (D) Autotrophs make their own food, but they cannot do so from nothing: they need carbon dioxide, water and light (plus chlorophyll) from their surroundings as raw materials and energy. So 'need nothing at all' is wrong. 'Completely independent' repeats the error. Autotrophs (green plants) do not eat small animals to 'get started' — they photosynthesise. They do not take ready-made food from the soil; that is the food-from-soil misconception. The accurate correction is that they still depend on CO₂, water and light.

56. (B) Carbon dioxide is a raw material of photosynthesis. The KOH removes it from one leaf's air, so that leaf cannot make starch and stays brown; only the normal-air leaf makes starch and turns blue-black. The student's error was treating CO₂ as something whose removal HELPS, when in fact CO₂ is essential and its removal stops food-making. 'Both turn blue-black, light alone is enough' ignores that CO₂ is needed too. 'Neither, destarching destroyed the ability' misunderstands destarching, which only empties stored starch and does not disable photosynthesis. 'KOH feeds the leaf' inverts KOH's role — it ABSORBS CO₂, it does not provide it.

57. (B) Zero NET oxygen exchange means the oxygen released by photosynthesis exactly matches the oxygen consumed by respiration at that instant — the compensation point, occurring around dawn and dusk as light changes. Both processes are running; their rates are equal. 'Stopped both processes' is wrong — respiration never stops, and at these moments photosynthesis is occurring too. 'Only respiring, using its own oxygen' describes night, not a zero-crossing where photosynthesis is balancing respiration. 'Photosynthesis stopped, respiration full speed' would give a net oxygen UPTAKE, not zero — that is night-time, not the balance point.

58. (D) The chapter's full picture is: green plants are autotrophs making food by photosynthesis; some organisms are heterotrophs feeding in special modes (parasitic like *Cuscuta*, insectivorous like the pitcher plant, saprotrophic like mushrooms, symbiotic like lichens and *Rhizobium*); and soil nitrogen is restored by nitrogen-fixing bacteria. Only the first option captures all of this. 'Food only at night' is false — photosynthesis needs light, so it occurs in the day. 'Plants and animals both make food from sunlight' is false — animals cannot photosynthesise. 'Food ready-made from the soil' is the central misconception the chapter overturns.

59. (A) A plant's body is largely made of food (glucose/starch and related substances) whose carbon comes from carbon dioxide fixed in the leaves during photosynthesis — so the great bulk of new solid mass is built from a gas in the air, not solid soil. Roots take up water and dissolved minerals, not whole solid soil that hardens into wood, so the first wrong option misdescribes how roots work. Oxygen is a PRODUCT released by photosynthesis, not a raw material the tree turns into solid wood. The girl's whole-soil idea is the food-from-soil misconception the chapter overturns: the trunk's mass traces mostly to carbon dioxide, with water supplying hydrogen.

60. (A) Water (carrying the dye) moves UP from the roots through the xylem, which is why the dye appears high in the leaves. Food made in the lit leaf (the labelled sugar) travels AWAY from the leaf — here downward toward the roots — through the phloem. So it is xylem for the dye, phloem for the sugar. The first wrong option swaps the two tissues' jobs. 'Xylem for both' is false — the plant has two distinct transport tissues, water in xylem and food in phloem. Stomata are leaf pores for gas exchange and chlorophyll is a pigment, not transport channels, so that option misnames the structures entirely.